Schmitt Trigger Based Level Detection and Noise Rejection Control Circuit

Course: CSE350

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***Abstract*—This paper presents the design, simulation, and hardware implementation of a Schmitt trigger based control circuit that performs automatic two-level switching of a lamp in response to ambient light intensity. The circuit uses a light dependent resistor (LDR) as the sensing element and an LM741 operational amplifier configured with positive feedback to establish two distinct switching thresholds. The lamp is energised when the sensed voltage rises above an upper threshold voltage V_{TH} and is de-energised only after the sensed voltage falls below a separate lower threshold voltage V_{TL}. The voltage difference between these two thresholds forms a hysteresis window that prevents repeated switching when the input signal fluctuates near the switching point. To establish a baseline for comparison, an ordinary single-threshold comparator was implemented on the same input signal. Experimental results confirm that the comparator produces unstable output near its threshold, whereas the Schmitt trigger produces clean, single-transition switching under identical conditions. The measured thresholds agree closely with the values predicted by analysis.**

***Index Terms*—Schmitt trigger, hysteresis, comparator, operational amplifier, level detection, noise rejection, LDR**

I. Introduction

An automatic lighting controller must energise a lamp when ambient illumination falls below a usable level and de-energise it once adequate illumination returns. A single fixed switching threshold is insufficient for this task, because any real sensor signal carries noise. When the sensed signal sits near the threshold, small fluctuations force the output to switch repeatedly between states. In a lighting circuit this appears as visible flicker, and the load is subjected to rapid and unnecessary switching cycles.

The Schmitt trigger addresses this problem by applying positive feedback around a comparator, producing two separate switching thresholds rather than one. The output changes state at the upper threshold V_{TH} when the input is rising, and returns at the lower threshold V_{TL} when the input is falling. Because $V_{TH} > V_{TL}$, any disturbance smaller than the difference between them cannot cause an unwanted transition.

The objectives of this work are:

1) To design and construct a Schmitt trigger circuit with two independently controllable switching thresholds;
2) To demonstrate two-level (hysteretic) control of a lamp, in which the switching level for turning the lamp on differs from the level for turning it off;
3) To demonstrate quantitatively that an ordinary comparator fails under the same noisy input conditions;
4) To verify the measured thresholds and hysteresis width against theoretical analysis.

II. Implementation

A. Theory of Operation

An operational amplifier operated without feedback acts as a comparator. If the non-inverting input exceeds the inverting input the output saturates at $+V_{sat}$; otherwise it saturates at $-V_{sat}$. Switching occurs at a single input level, so a noisy input crossing that level produces multiple output transitions.

In the Schmitt trigger a fraction of the output is returned to the *non-inverting* input through a resistive divider formed by R_1 and R_2. This positive feedback shifts the effective threshold in a direction that opposes any further change, so the switching level depends on the current output state. Defining the feedback factor

$$k = \frac{R_1}{R_2}, \tag{1}$$

the two thresholds referred to the input node are

$$V_{TH} = V_{ref}(1 + k) + k\,V_{sat}, \tag{2}$$

$$V_{TL} = V_{ref}(1 + k) - k\,V_{sat}. \tag{3}$$

The hysteresis width follows directly as

$$\Delta V = V_{TH} - V_{TL} = 2\,k\,V_{sat}. \tag{4}$$

Equations (2)–(4) show that the two design parameters are independent. The *width* of the hysteresis window is set only by the resistor ratio k and the saturation voltage, while the *position* of the window is set only by the reference voltage V_{ref}. This separation is what allows the circuit to be adapted to a different application without redesigning the feedback network.

B. Circuit Design

The complete circuit is shown in Fig. 1. It operates from a dual ± 9 V supply formed by two 9 V batteries connected in series, with the common node taken as the circuit ground. Three LM741 operational amplifiers are used.

Sensing stage. A light dependent resistor forms the lower arm of a potential divider, with a fixed 2.2 kΩ resistor in series with a 10 kΩ trimmer forming the upper arm. The trimmer permits calibration against the ambient lighting of the test environment. A 22 kΩ resistor is placed in parallel

with the LDR. This resistor is essential: in darkness the LDR resistance rises into the hundreds of kilohms, which would drive the divider output close to the positive rail and beyond the specified input common-mode range of the LM741. The parallel resistor bounds the divider output to approximately 6 V, keeping the op-amp inputs within their valid operating range under all lighting conditions.

Buffer stage (U1). The sensing divider is followed by a unity-gain voltage follower. Without this buffer, the current drawn by the feedback network of the Schmitt trigger would load the divider and displace the switching thresholds from their designed values. The buffer presents a very high input impedance to the divider and a very low output impedance to the following stages, so the measured thresholds agree with the analysis.

Comparator stage (U2). An LM741 is operated open-loop with the sensor voltage on the non-inverting input and V_{ref} on the inverting input. This stage provides the single-threshold reference behaviour against which the Schmitt trigger is compared.

Schmitt trigger stage (U3). $R_1 = 10$ kΩ connects the buffered sensor voltage to the non-inverting input, and $R_2 = 100$ kΩ returns the output to the same node, establishing positive feedback. V_{ref} is applied to the inverting input.

Output stage. The Schmitt trigger output drives a BC547 transistor through a 1 kΩ base resistor and a 1N4148 diode. The diode blocks reverse base current when the op-amp output swings to $-V_{sat}$, which would otherwise damage the transistor. The transistor switches a white LED, which represents the controlled lamp.

Each op-amp output additionally drives an anti-parallel red and green LED pair, allowing the state of each stage to be observed directly. The anti-parallel arrangement is necessary because the op-amp output is bipolar.

C. Mathematical Analysis

The reference voltage is generated by a divider of $R = 12$ kΩ and 10 kΩ across the $+9$ V rail:

$$V_{ref} = 9 \times \frac{10}{12 + 10} = 4.09 \text{ V}. \tag{5}$$

With $R_1 = 10$ kΩ and $R_2 = 100$ kΩ, the feedback factor is

$$k = \frac{R_1}{R_2} = \frac{10}{100} = 0.1. \tag{6}$$

The output saturation voltage was measured at the op-amp output as $V_{sat} = 7.5$ V.

Substituting into (2)–(4):

$$V_{TH} = 4.09(1.1) + 0.1(7.5) = 4.50 + 0.75 = 5.25 \text{ V}, \tag{7}$$

$$V_{TL} = 4.09(1.1) - 0.1(7.5) = 4.50 - 0.75 = 3.75 \text{ V}, \tag{8}$$

$$\Delta V = 2(0.1)(7.5) = 1.50 \text{ V}. \tag{9}$$

The sensing network was calibrated so that the buffered sensor voltage spans approximately 0.4 V in bright illumination to 6.0 V in darkness. Both thresholds therefore lie within the achievable range with adequate margin at each end, which is a necessary condition for the circuit to switch in both directions.

D. Simulation

The circuit was first constructed and verified in Tinkercad. The simulation confirmed correct polarity of the sensing network (rising voltage with decreasing illumination), correct saturation of all three op-amp stages, and the presence of two distinct switching points on the Schmitt trigger stage.

A discrepancy was observed between the simulated and measured hysteresis width. The simulator's operational amplifier model saturates much closer to the supply rails than a physical LM741, giving $V_{sat} \approx 9$ V rather than the measured 7.5 V. By (4) this predicts a simulated window of 1.8 V against a measured window of 1.5 V. This is not an error but a direct consequence of the model: substituting the appropriate V_{sat} into (4) reproduces each result correctly. The observation confirms experimentally that the hysteresis width depends on the actual output swing of the device rather than on the supply voltage.

E. Hardware Implementation

The circuit was assembled on a breadboard using three LM741 operational amplifiers, a BC547 transistor, an LDR, a 10 kΩ trimmer for calibration, a 10 kΩ potentiometer for controlled input sweeping, and discrete passive components. Construction proceeded stage by stage, with each stage verified before the next was added.

A potentiometer was included as an alternative input source. A single jumper selects whether the buffer is fed from the LDR or from the potentiometer. The potentiometer provides a smooth, low-noise, manually controlled input, which allows the switching thresholds to be located precisely with a multimeter. The LDR provides a genuinely noisy real-world input, which is required for the comparator instability demonstration. Because both sources feed the same buffer, the measured thresholds are identical for either source, confirming that the thresholds are a property of the circuit rather than of the sensor.

F. Experimental Results

1) Threshold Measurement: The buffered sensor voltage V_{sense} was monitored with a digital multimeter while the input was swept slowly using the potentiometer. The voltage was recorded at the instant the lamp changed state in each direction. Sweeping upwards, the lamp switched on as V_{sense} passed V_{TH}; sweeping downwards, it switched off only after V_{sense} had fallen to V_{TL}. The difference between the two recorded values gives the hysteresis width directly.

2) Comparison with an Ordinary Comparator: With the LDR selected as the input source and the ambient illumination adjusted so that V_{sense} rested close to V_{ref}, the comparator output was observed to switch repeatedly between states. The Schmitt trigger, driven from the same buffered signal at the same instant, remained in a single stable state. The lamp driven by the Schmitt trigger showed no flicker.

3) Hysteresis Characteristic: The transfer characteristic was captured on an oscilloscope in X–Y mode, with V_{sense} applied to the horizontal axis and the Schmitt trigger output to the vertical axis. The resulting trace forms a closed

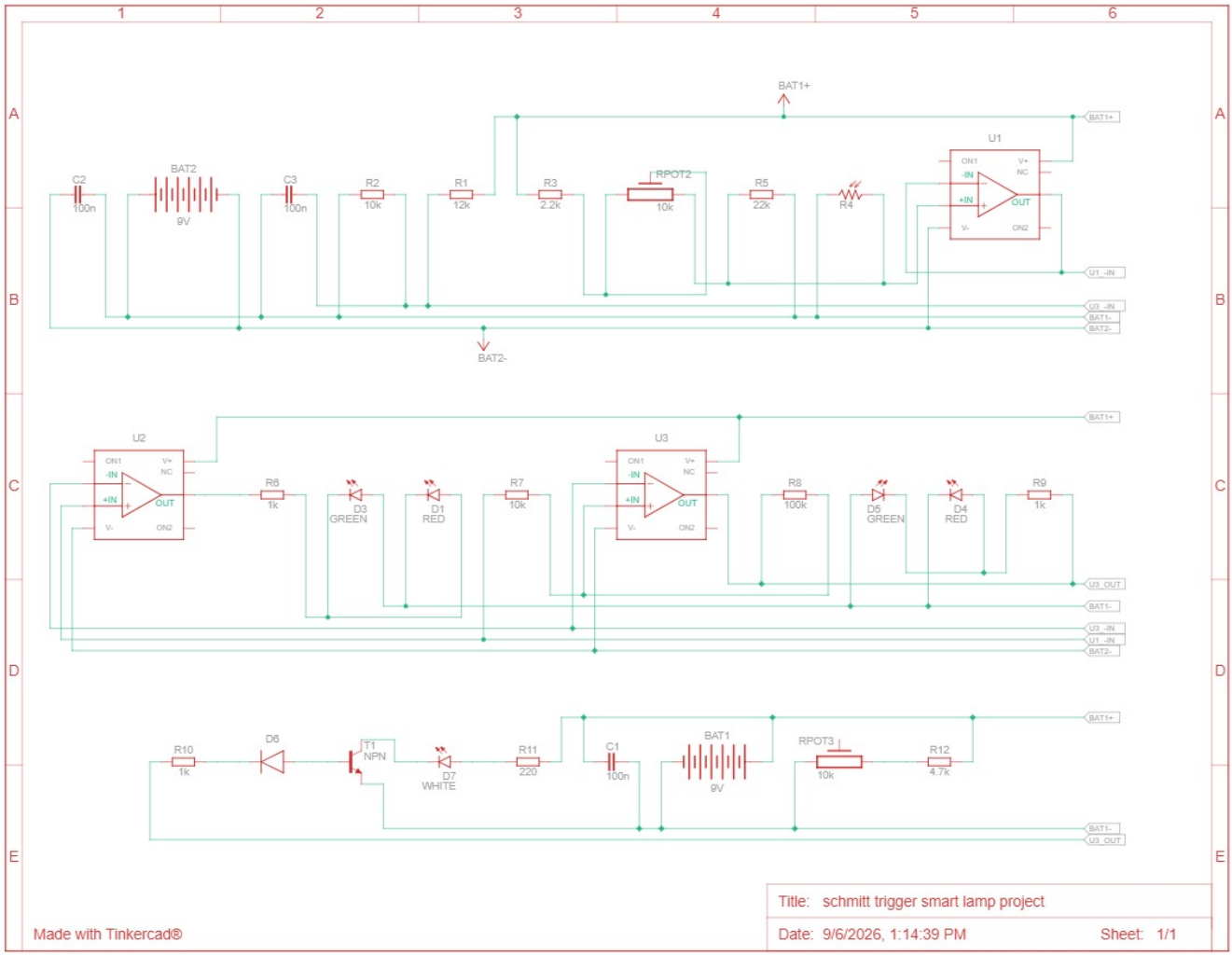


Fig. 1. Complete circuit schematic of the Schmitt trigger based level detection and noise rejection control circuit.

rectangular loop whose horizontal width corresponds directly to the hysteresis window. Under the same measurement the ordinary comparator produces a single vertical transition with no enclosed area, confirming the absence of hysteresis.

G. Challenges Faced

Several practical difficulties were encountered and resolved during implementation.

Insufficient sensor headroom. In the initial calibration the sensing network was adjusted so that the divider output reached only 4.5 V in complete darkness. Since the calculated upper threshold is 5.25 V, the input could never reach the switching point and the lamp would not turn on under any lighting condition. The upper arm resistance was reduced so that the dark reading rose to approximately 6.0 V, placing both thresholds comfortably within the achievable range while remaining below the input common-mode limit of the LM741.

Operational amplifier input range violation. Early testing showed erratic behaviour when the LDR was fully covered. The cause was traced to the divider output approaching the positive supply rail as the LDR resistance rose into the megohm region, exceeding the LM741 input common-mode range. A 22 kΩ resistor placed in parallel with the LDR bounds the maximum divider output and eliminated the problem.

Threshold displacement due to loading. When the sensing divider was connected directly to the Schmitt trigger, the current drawn by the feedback network altered the divider output and the measured thresholds departed from their calculated values. Inserting a unity-gain buffer between the sensor and the trigger isolated the two networks and restored agreement with the analysis.

Interpretation of output node voltage. During testing, the voltage at the LED node was measured as approximately 1.9 V while the output was in its high state, which initially appeared to indicate a fault. This is in fact the expected forward voltage of the LED, with the remaining supply voltage dropped across the series 1 kΩ resistor. Diagnostic measurements were subsequently taken at the op-amp output terminal rather than at the indicator node.

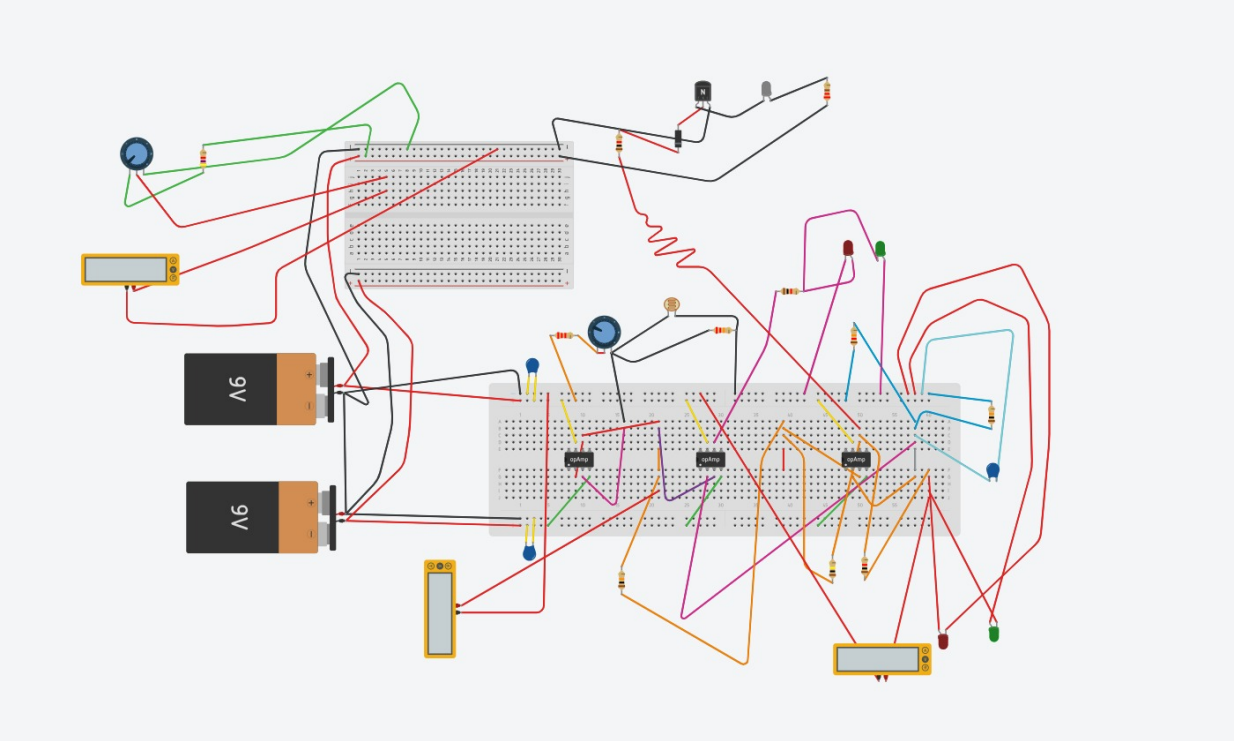


Fig. 2. Tinkercad simulation of the complete circuit.

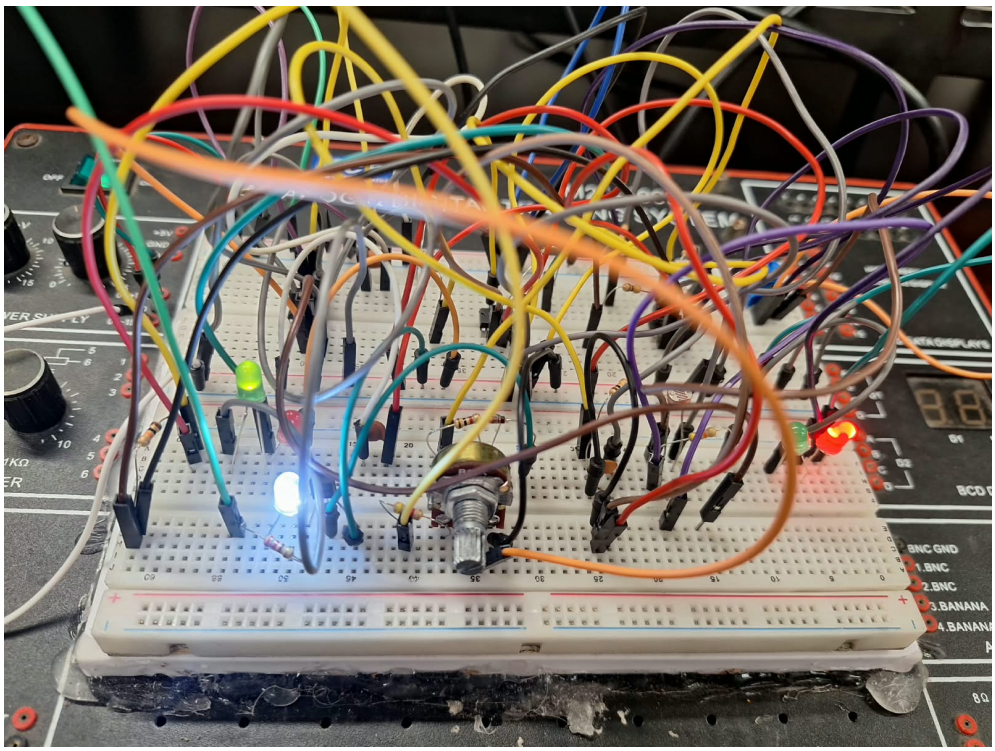


Fig. 3. Breadboard implementation showing the sensing network, buffer, comparator, Schmitt trigger, and lamp driver stages.

Distinguishing genuine hysteresis from apparent behaviour. Because the resistance of an LDR varies over several orders of magnitude with illumination, the hysteresis window occupies only a small portion of the physical range of hand movement, and is not easily perceived by eye. Substituting a potentiometer for the sensor produced a linear and controllable

input sweep, allowing both thresholds to be located and recorded accurately.

III. Conclusion

A Schmitt trigger based two-level control circuit was designed, simulated, and implemented in hardware. The circuit switches a lamp on when the sensed voltage rises above V_{TH} and off only after it falls below V_{TL}, so that the two switching levels are deliberately separated. The measured thresholds and hysteresis width agree with the values predicted by analysis. A direct comparison against an ordinary comparator driven from the same input confirmed that the single-threshold circuit switches repeatedly under noisy conditions while the Schmitt trigger does not.

The work also demonstrated that the two design parameters of the circuit are independent: the hysteresis width is set by the feedback resistor ratio, while the position of the window is set by the reference voltage. Either may be adjusted without disturbing the other.

Several extensions are possible. Replacing the LM741 with a rail-to-rail comparator such as the LM339 would allow single-supply operation and remove the need for a dual rail. Replacing the transistor with a relay or MOSFET driver would allow the same control law to switch a mains-powered lamp. Introducing a digitally controlled potentiometer in place of the reference divider would permit the switching levels to be set programmatically. Finally, adding a second Schmitt trigger stage with a different reference would allow multi-level control, for example dimming the lamp at one illumination level and switching it fully on at a lower one.

References

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